

College of Industrial Technology  
King Mongkut's University of Technology North Bangkok



Midterm Examination of semester 2

Year: 2017

Subject: 394172 Mathematics 2

Section: 15-18

Date : 6 March 2018

Time: 08.00-10.00

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Class: \_\_\_\_\_

**Instructions:**

1. The examination has 10 pages (including this page) and a total score of 100 points.
2. Write your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor permission.
6. You not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculations are NOT allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

**Cheating in the exam is considered an extremely serious  
offence which will result in expulsion from the University.**

**Question 1** Let  $p, q, r$  be statements

1.1 Given statement  $[(p \rightarrow \sim r) \wedge q] \rightarrow (p \rightarrow \sim q)$  is false. Determine the truth value of the statement  $\sim r \rightarrow (p \rightarrow \sim q)$ . (5 points)

1.2 Given statement  $(\sim p \rightarrow \sim r)$  is false. Determine the truth value of the statement  $q \rightarrow \sim [(p \vee (q \rightarrow r))]$ . (5 points)

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Question 2 2.1 Which the following choices that is equivalent to the statement  $\sim p \rightarrow (q \rightarrow (\sim r \rightarrow p))$ . (5 points)

1.  $p \vee \sim q \vee r$
3.  $(p \rightarrow q) \wedge r$

2.  $p \rightarrow (\sim q \vee r)$
4.  $(p \rightarrow q) \vee \sim r$

2.2 Show whether the statement  $[((p \wedge \sim q) \rightarrow r) \wedge (\sim q \wedge p)] \rightarrow (\sim r \rightarrow \sim p)$  is tautology or not. (5 points)

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**Question 3** Given  $x, y, z \in \mathbb{R}$  which are real numbers. Find the truth value of the following quantified statements.

3.1  $\exists x \left[ \sqrt{x} > |x| \right] \rightarrow \forall x \left[ \frac{0}{x^2} = 0 \right]$ . (5 points)

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3.2  $\exists x \exists y \left[ x^2 < \frac{x}{y} \right] \rightarrow \forall x \exists y \exists z \left[ z^3 = xy \right]$ . (5 points)

**Question 4** 4.1 Find the negation of the quantified statement  $\forall x \exists y \left[ x^2 < xy \rightarrow \frac{y}{x} > 1 \right]$ .

(5 points)

4.2 Given the arguments as follows:

- Hypothesis
1.  $(p \wedge \neg q) \rightarrow r$
  2.  $q \rightarrow s$
  3.  $\sim r$
  4.  $p$

Conclusion:  $s$ .

Determine whether this argument conclusion is valid or not. (5 points)

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**Question 5** 5.1 Given  $x = (E5FAC)_{16}$ ,  $y = (A78DA)_{16}$  and  $z = (DF78B)_{16}$ , find the expression of  $x - y + z$  in base 16. (5 points)

5.2 Convert the number  $236127_8$  to a number in base 10. (5 points)

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Question 6 6.1 Find the product of  $(ABCD)_{14}$  and  $(BA)_{14}$ . (5 points)

6.2 Find the value of  $a + b$  of the expression  $(0.45)_{10} = (0.2a11b...)_{6}$ . (5 points)

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**Question 7** 7.1 Convert each of the following numbers  $(0.235)_6$  and  $(2.812)_{10}$  to numbers in base 4 expansion. (5 points)

7.2 Find the result of  $[12321_4 - 567_8] \times AC_{16}$  and answer your result in base 16.  
(5 points)

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**Question 8** Find sum of all solutions of the algebraic equation

$$1 + \frac{2}{(2x^2 - x)^2} = \frac{3}{x(2x - 1)}. \quad (10 \text{ points})$$

**Question 9** Given a system of second-degree equations with two variables

$$xy + y^2 = 24$$

$$x^2 + 2xy + 10y^2 = 145$$

Find a set of the solutions. (10 points)

**Question 10** Find all solutions of the system of nonlinear equations (10 points)

$$x^2 - yz = 64,$$

$$y^2 - xz = 88,$$

$$z^2 - xy = 4.$$

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